Ex. 19. ***Which digits should come in place of * and $ if the number 62684*$ is divisible by both 8 and 5?***

Sol. Since the given number is divisible by 5, so 0 or 5 must come in place of $. But, a number ending with 5 is never divisible by 8. So, 0 will replace $.

Now, the number formed by the last three digits is 4*0, which becomes divisible by 8, if * is replaced by 4.

Hence, digits in place of * and $ are 4 and 0 respectively.

Ex. 20. ***Show that 4832718 is divisible by 11.***

Sol. (Sum of digits at odd places) – (Sum of digits at even places) = (8 + 7 + 3 + 4) – (1 + 2 + 8) = 11, which is divisible by 11.

Hence, 4832718 is divisible by 11.

Ex. 21. ***Is 52563744 divisible by 24?***

Sol. $24 = 3 \times 8$, where 3 and 8 are co-primes.

The sum of the digits in the given number is 36, which is divisible by 3. So, the given number is divisible by 3.

The number formed by the last 3 digits of the given number is 744, which is divisible by 8. So, the given number is divisible by 8. Thus, the given number is divisible by both 3 and 8, where 3 and 8 are co-primes. So, it is divisible by 3×8, i.e. 24.

Ex. 22. ***What are the values of M and N respectively if M39048458N is divisible by both 8 and 11, where M and N are single-digit integers?***

Sol. Since the given number is divisible by 8, it is obvious that the number formed by the last three digits, i.e. 58N is divisible by 8, which is possible only when $N = 4$.

Now, (sum of digits at even places) – (sum of digits at odd places)

$= (8 + 4 + 4 + 9 + M) - (4 + 5 + 8 + 0 + 3)$

$= (25 + M) - 20 = M + 5$, which must be divisible by 11.

So, $M = 6$.

Hence, $M = 6$, $N = 4$.

Ex. 23. ***Find the number of digits in the smallest number which is made up of digits 1 and 0 only and is divisible by 225.***

Sol. $225 = 9 \times 25$, where 9 and 25 are co-primes.

Clearly, a number is divisible by 225 if it is divisible by both 9 and 25.

Now, a number is divisible by 9 if the sum of its digits is divisible by 9 and a number is divisible by 25 if the number formed by the last two digits is divisible by 25.

$\therefore$ The smallest number which is made up of digits 1 and 0 and divisible by 225 = 11111111100.

Hence, number of digits = 11.

Ex. 24. ***If the number 3422213pq is divisible by 99, find the missing digits p and q.***

Sol. $99 = 9 \times 11$, where 9 and 11 are co-primes.

Clearly, a number is divisible by 99 if it is divisible by both 9 and 11.

Since the number is divisible by 9, we have:

$(3 + 4 + 2 + 2 + 2 + 1 + 3 + p + q)$ = a multiple of 9

$\Rightarrow 17 + (p + q) = 18$ or 27

$\Rightarrow p + q = 1$ ...(*i*) or $p + q = 10$ ...(*ii*)

Since the number is divisible by 11, we have:

$(q + 3 + 2 + 2 + 3) - (p + 1 + 2 + 4) = 0$ or a multiple of 11

$\Rightarrow (10 + q) - (7 + p) = 0$ or 11

$\Rightarrow 3 + (q - p) = 0$ or 11

$\Rightarrow q - p = -3$ or $q - p = 8$

$\Rightarrow p - q = 3$ ...(*iii*) or $q - p = 8$ ...(*iv*)

Clearly, if (*i*) holds, then neither (*iii*) nor (*iv*) holds. So, (*i*) does not hold.

Also, solving (*ii*) and (*iii*) together, we get: $p = 6.5$, which is not possible.

Solving (*ii*) and (*iv*) together, we get: $p = 1$, $q = 9$.

Ex. 25. ***x is a positive integer such that $x^2 + 12$ is exactly divisible by x. Find all the possible values of x.***

Sol. $\dfrac{x^2+12}{x} = \dfrac{x^2}{x} + \dfrac{12}{x} = x + \dfrac{12}{x}$.

Clearly, 12 must be completely divisible by x.

So, the possible values of x are 1, 2, 3, 4, 6 and 12.